



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1

2024/2025

SECR1213-02 NETWORK COMMUNICATIONS

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1.0 BRIEF EXPLANATION ABOUT THE TASK

The task involves creating a group for a university project that can create a new building plan for the Faculty of Computing (FC) in order to accommodate future growth and learning requirements. Our four-person team, Y-Link, is made up of Dayang, Farra, Harini and Safiya. The team's objective is to understand the case study and provide a cost-effective, a "ready for future" setting that improves network security, teaching and learning.

The case study outlines the goal of the task of creating a readable and clear floor plan that efficiently accommodates the required areas. The floor plan is about a 2-storey building that includes 4 labs which includes 2 general-purpose labs, 1 Cisco Network Lab and 1 Embedded lab for IoT and sensors), a video conferencing room, a hybrid classroom and a student lounge with Wi-Fi. High-speed internet will be available in every 14m x 10m lab to enable instruction in accordance with the 4th Industrial Revolution. The design will prioritize security, reliable high-performance networking and seamless learning in addition to future scalability and advanced teaching capabilities.

2.0 GROUND FLOOR PLAN

Figure 1 is a comprehensive design for a facility that includes multiple functional areas, all interconnected by a central corridor. The design consists of rooms tailored for different purposes, creating a well-organized and accessible floor plan. The Ground Floor consists of a Video Conference Room, Student Lounge, Hybrid Classroom, Prayer Rooms, Staff Room, Server Room and Restrooms. The layout also includes accessible corridors and entrances which connect all rooms and ensure easy access throughout the facility. Lastly, the floor is equipped with an elevator at the left side of the layout. Figure 2 shows the indicator of the ground floor.

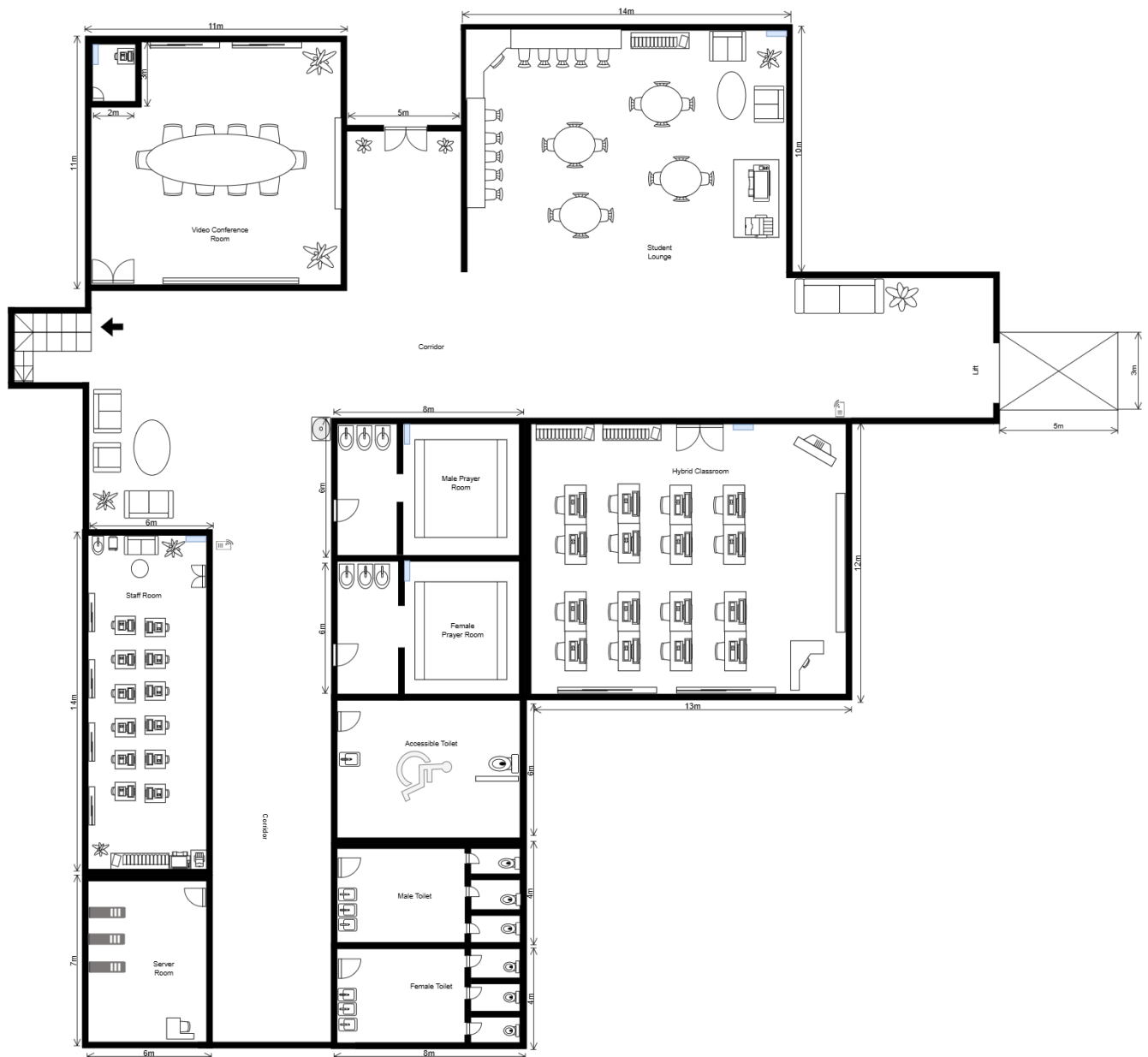


Figure 1 : Layout of Ground Floor

Indicators :	
	- Desk (75cm x 50cm), (475cm x 75cm), (325cm x 200cm)
	- Computer (50cm x 40cm)
	- Chair (50cm x 25cm)
	- Window (180cm x 75cm)
	- Tv (250cm x 90cm)
	- Whiteboard (500cm x 25cm)
	- Door (70cm x 125cm)
	- Toilet (50cm x 75cm)
	- Flowerpot (50cm x 50cm), (100cm x 100cm)
	- Sink (125cm x 75cm)
	- WiFi
	- Refrigerator (50cm x 75cm)
	- Switch
	- Lift
	- Printer (60cm x 70cm)
	- Stairs
	- Gliding window (300cm x 50cm)
	- Conference table (625cm x 225cm)
	- Corner desk (200cm x 300cm)
	- Bookcase (260cm x 75cm)
	- Sofa (150cm x 125cm), (375cm x 150cm)
	- Table (200cm x 100cm), (240cm x 160cm)
	- Table (150cm x 150cm)
	- Copier (150cm x 75cm), (100cm x 75cm)
	- Prayer room
	- Sink (50cm x 75cm)
	- Sink (60cm x 80cm)
	- Sofa (125cm x 125cm)
	- Server
	- Corner desk (100cm x 125cm)

Figure 2 : Indicator for Ground Floor

1. Video Conference Room

Figure 3 shows a spacious, rectangular video conference room measuring 11 meters by 11 meters. The central feature is a large, oval conference table with six chairs arranged around it. Along one wall, there are two windows, which is to provide natural light to the room. Near the top-left corner, there is a small, enclosed section that appears to be a technician room, measuring 2 meters by 5 meters. It has a door for access, and inside, there's a workstation for IT staff. The main entrance to the video conference room is a double door located in the bottom-left corner, providing a wide entryway.

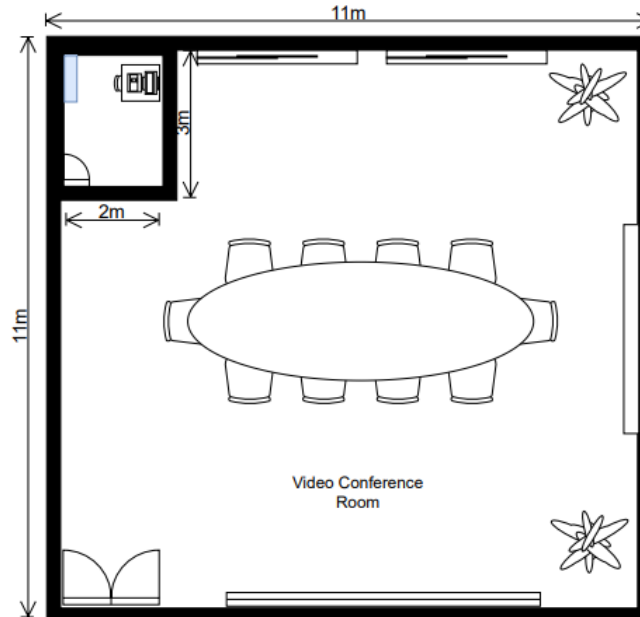


Figure 3 : Layout of Video Conference Room

2. Student Lounge

Figure 4 shows the student lounge design which is a comfortable, multifunctional space with various seating and activity areas, measuring 14 meters by 10 meters. The left side of the room features a counter area providing a space for casual seating or studying. Near the top, there is a shelf or small storage unit for books or personal belongings, encouraging a study-friendly environment. In the center of the room, several round tables are spread out, each with four chairs, creating group seating areas. The right side of the room features a cozy seating area with two sofas facing each other and a small oval coffee table in between. The student lounge also offers a printer and copier.

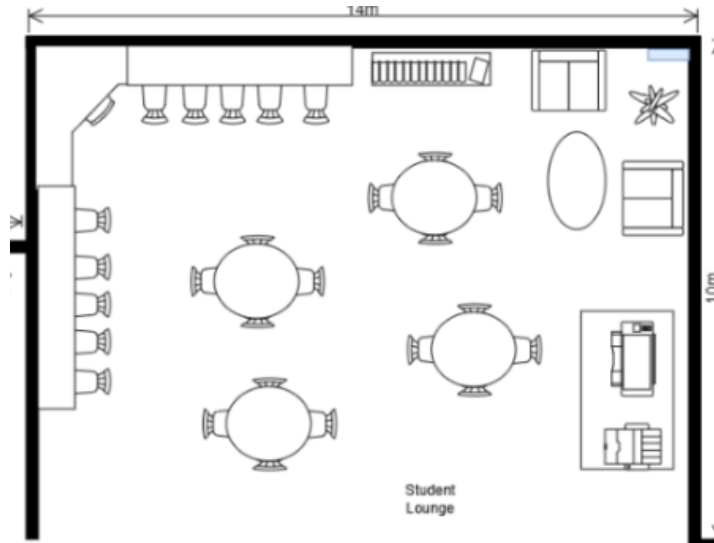


Figure 4 : Layout of Student Lounge

3. Hybrid Classroom

Figure 5 shows the hybrid classroom design which accommodates both in-person and remote learning, measuring 13 meters by 12 meters. At the front of the room, there is a lecturer's table positioned slightly to the right. Facing the table, six rows of student desks are arranged in pairs, with each desk having a chair and computer. Along the top wall, a set of shelves for books, teaching materials, or student resources, making it easy to access necessary supplies. The room also features a large double door entry in the center of the top wall and equipped with a flat tv on the right side. Lastly, the back wall has two long windows, allowing natural light to enter.

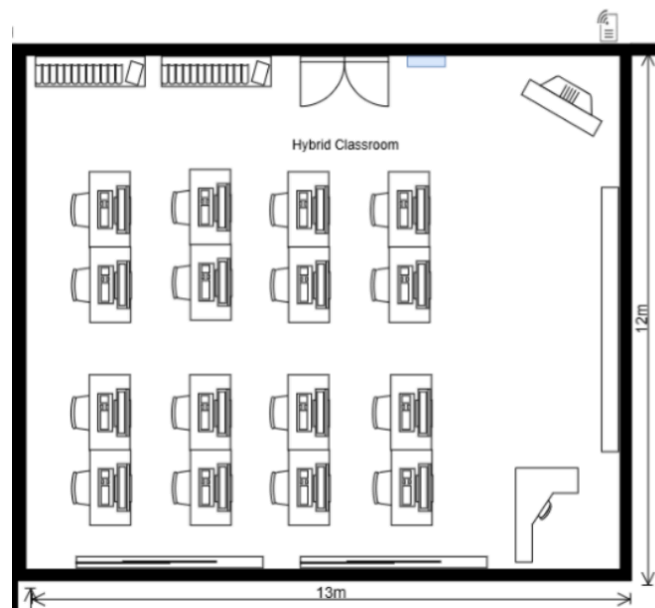


Figure 5 : Layout of Hybrid Classroom

4. Prayer Room

Figure 6 shows the prayer room layout which consists of two equal sections, each measuring 6 meters by 8 meters, with separate areas for male and female users. Each section has its own entrance, located on the left side, leading into a dedicated area with three washing stations for ablution. Adjacent to the washing stations, there is a doorless passage that opens into the main prayer area.

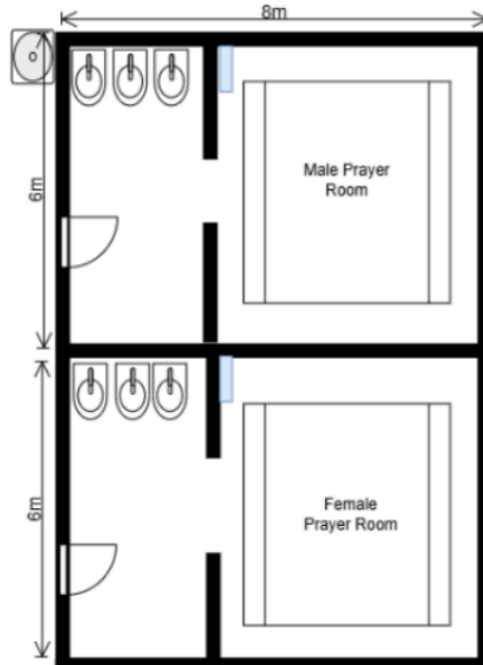


Figure 6 : Layout of Prayer Room

5. Toilet

Figure 7 shows the toilet layout which is designed to accommodate male, female, and accessible needs within a structured 8-meter-wide area. At the front, there is a spacious accessible toilet, measuring approximately 6 meters by 4 meters, designed to be shared between male and female users. This room provides ample space for a wheelchair to move freely and includes an accessible toilet and grab bars for added support.

Beside the toilet are separate male and female toilet facilities, each measuring about 4 meters by 4 meters. The male toilet area has three sinks positioned along the left wall, with three private stalls. Similarly, the female toilet includes three private stalls aligned on one side, allowing multiple users to use the space simultaneously. Each section has a designated entrance with no crossover between the male and female facilities.

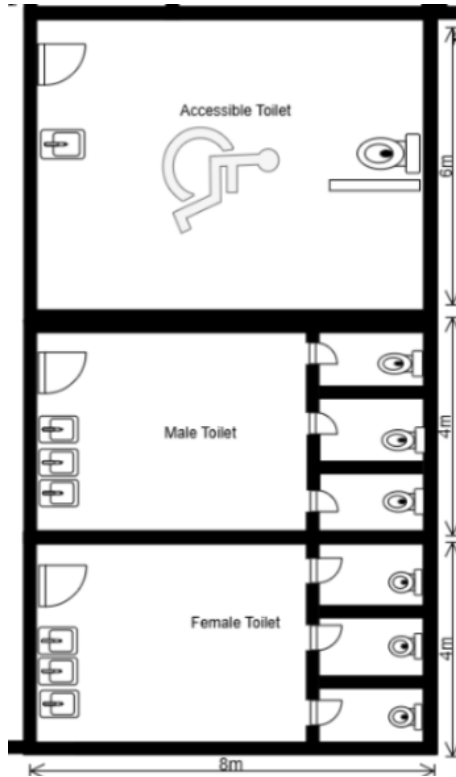


Figure 7 : Layout of Toilet

6. Server Room

Figure 8 shows the server room layout which is a compact, rectangular space measuring 6 meters by 7 meters, specifically designed to house essential IT equipment. Three server racks are aligned along the left wall, providing ample space for organizing servers, networking equipment, and other necessary hardware. The room has a single door in the upper-right corner for controlled access.

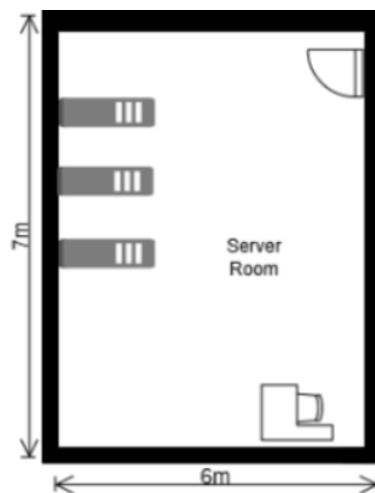


Figure 8 : Layout of Server Room

7. Staff room

Figure 9 shows the staff room layout which is designed within a 6-meter by 14-meter rectangular space. Upon entering through the door at the top right corner, there is a small seating area equipped with a couch, a coffee table, refrigerator and a plant, creating a comfortable spot for relaxation. Adjacent to this seating area is a sink for convenience. In the center of the room, there are two rows of desks, each containing six individual workstations arranged back-to-back, totaling twelve workspaces. Along the left wall, there are two windows, which provide natural light to the room. At the end of the room, there is additional storage space, including a filing cabinet, and a printer.

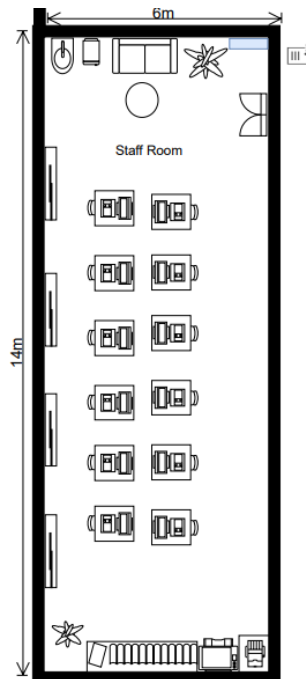


Figure 9 : Layout of Staff Room

Figure 10 shows the first floor that consists of two large General Purpose Labs, a Cisco Network Lab, and an Embedded Lab, each equipped with desks and computers for student use. The layout also includes supporting facilities such as a Staff Room and a Pantry. Centrally, there are Male and Female Toilets, along with an Accessible Toilet. The layout is organized around a central corridor for easy navigation, and an elevator located near the lower right corner provides access to other floors. Lastly, the floor is equipped with an elevator on the left side similar to the one on the Ground Floor. Figure 11 shows the indicator for the first floor.

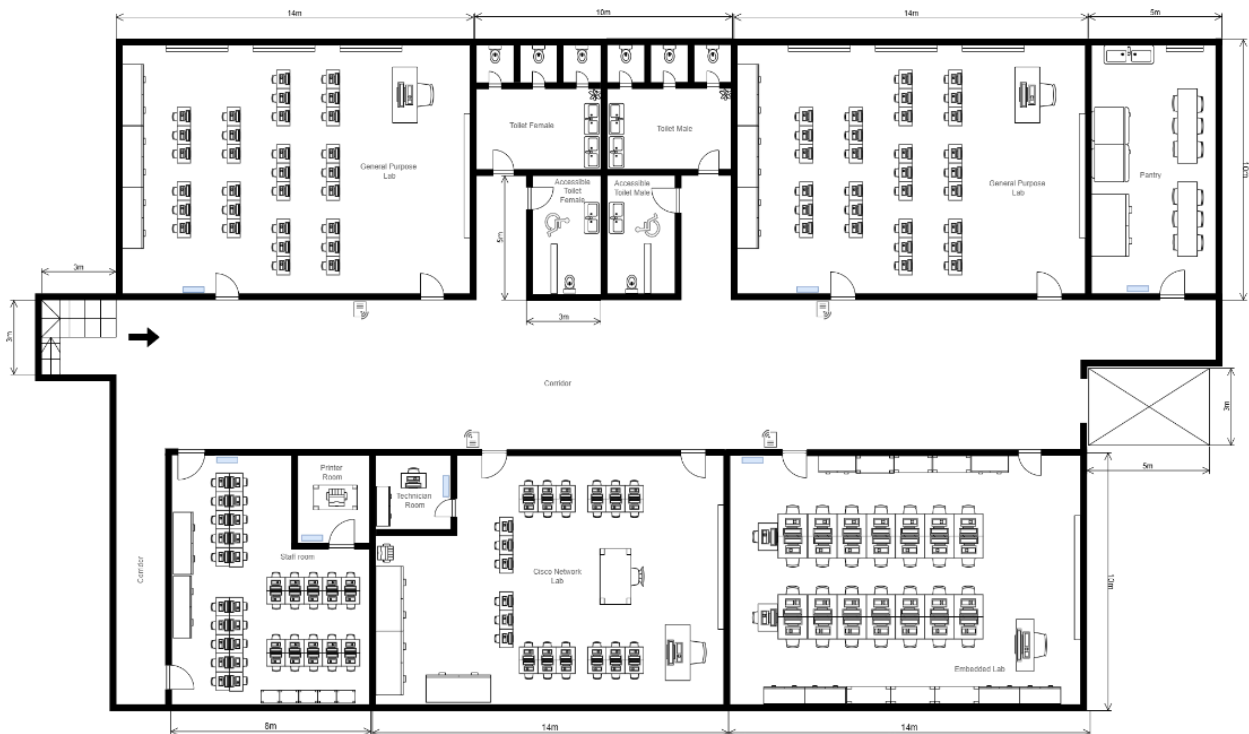


Figure 10 : Layout of First Floor

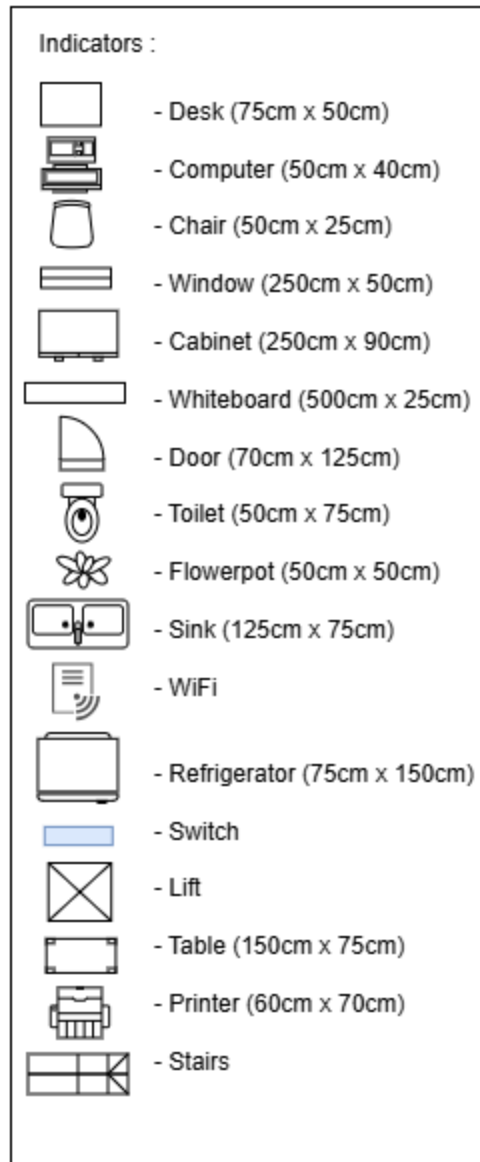


Figure 11 : Indicator for first floor

1. General Purpose Lab (2 identical labs)

Figure 12 shows the general-purpose lab layout which includes multiple rows of desks with computers, specifically 30 workstations arranged in a way that facilitates both individual work and collaborative activities. Each desk is equipped with a chair and is positioned to allow easy movement and access throughout the lab. Additionally, there is a dedicated instructor's area located at the front, featuring a larger desk and equipment for presentation purposes. Furthermore, the lab is equipped with high-speed internet connections to support education in line with the Fourth Industrial Revolution (4IR).



Figure 12 : Layout of 2 General Purpose Labs

2. Embedded Lab

Figure 13 shows the Embedded Lab with a design to support hands-on learning with a layout that accommodates various devices and peripherals essential for education in embedded systems. Spanning 14 meters by 10 meters, the room provides ample space for multiple users. Inside, four rows of workstations are arranged neatly, each equipped with desks and chairs. At the side of the room, a dedicated equipment area can be used to house critical networking hardware such as routers, switches, and servers, essential for hands-on practice and simulations. With two doors positioned at opposite ends, the lab ensures smooth entry and exit.

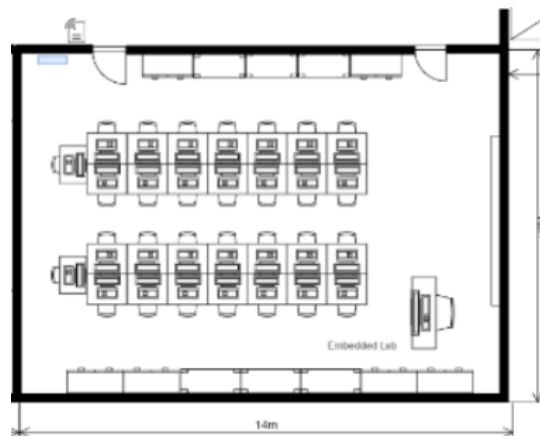


Figure 13 : Layout of Embedded Lab

3. Cisco Network Lab

Figure 14 shows the Cisco network lab thoughtfully designed to facilitate practical networking exercises and collaborative work. Measuring 14 meters by 10 meters, the lab features multiple rows of workstations arranged around the perimeter and center. Each

workstation includes a computer and ample desk space for students to set up and connect peripherals or embedded devices.

At the center of the room is a dedicated area with a larger work table and monitor, ideal for demonstrations or group projects. The instructor's desk is positioned at the front of the lab, equipped with additional presentation tools. A separate equipment storage area is located at the bottom left. A technician room is located at the top left, ensuring technical support is available if needed. High-speed internet and connectivity options are integrated to enhance learning experiences aligned with industry standards, creating an organized and versatile environment for education.

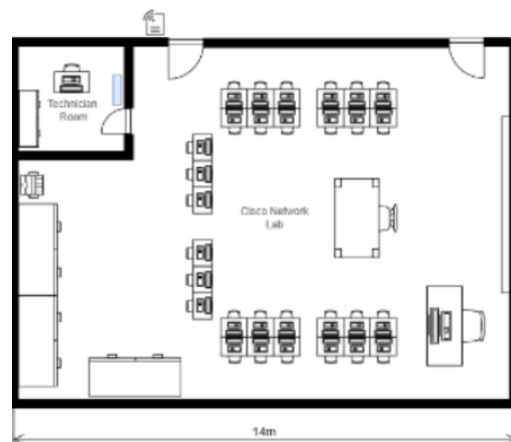


Figure 14 : Layout of Cisco Network Lab

4. Toilet (Male and Female)

Figure 15 shows the toilet facility layout purposely designed to accommodate both male and female users, with dedicated sections for each. The overall structure measures 10 meters in width and 5 meters in depth. Each section has multiple stalls, with both having three toilet stalls. Both sections have a central handwashing area with multiple sinks. Additionally, accessible toilets are provided for both genders, each measuring 3 meters by 5 meters and equipped with fixtures and space to support users with mobility challenges.

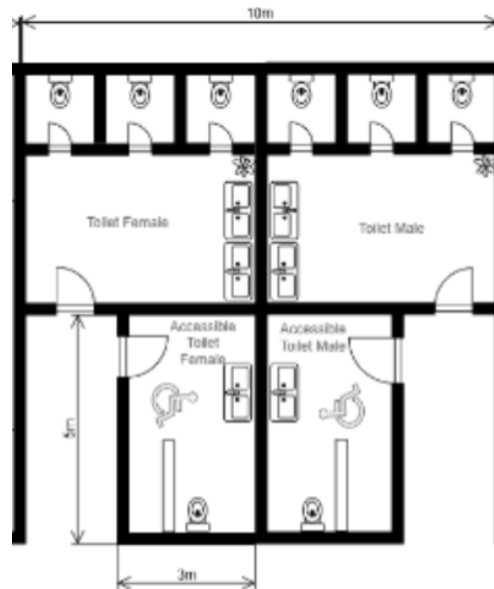


Figure 15 : Layout of Toilet (Male and Female)

5. Staff Room

Figure 16 shows the staff room layout which is designed to provide a functional workspace for staff, complete with a dedicated printer room for convenient document handling. The main area of the staff room includes multiple workstations arranged in rows, each with ample desk space and seating. Adjacent to these workstations is a set of storage cabinets along the wall for organizing materials and resources. The printer room is a separate, enclosed space located within the staff area, featuring a central printer/copier unit.



Figure 16 : Layout of Staff Room

6. Pantry

Figure 17 shows the pantry layout designed in a 5-meter by 10-meter rectangular space. On the left side, there is a designated area with a sink and countertop, as well as two large refrigerators placed side by side to provide ample storage. The central area is arranged with two long dining tables, each equipped with seating for four people on each side, allowing up to eight people to sit and eat together comfortably.

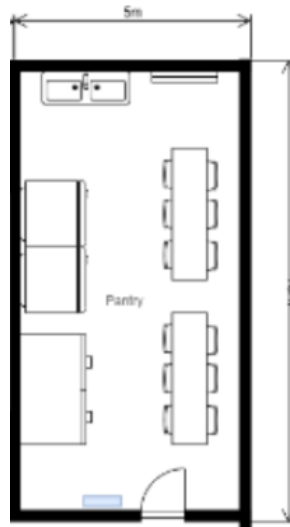


Figure 17 : Layout of Pantry

4.0 MEETING MINUTES

Date/Time		29 Oct 2024, 9.50 p.m.	
Location		Online (Google Meet)	
Agenda		1. Discuss the questions on Task 1	
		2. Way to arrange the floor plan	
		3. Tools for designing the floor plan	
		4. Task distribution for each person	
Meeting Host		Dayang Farah Farzana	
Attendance			
Name		Time	Reason For Absence
Dayang Farah Farzana		9.50	-
Farra Nurzahin		9.50	-
Harini		9.50	-
Safiya Nursyahadah		9.50	-
Minutes			
No.	Item Discussed	Details	Person in charge
1.	A short discussion on question given on Task 1	- All members read the question and rubric together.	All members
2.	Way to arrange the floor plan	- Safiya suggested that all the labs need to be on the same floor to reduce the cost. - All members agreed to the suggested plan.	Safiya

3.	Tools for designing the floor plan.	- Harini recommends Drawio and Floorplancreator to be our design tools. - Farra discovers that some features on Drawio more convenient. - All members agreed to use Drawio.	Farra, Harini
4.	Task distribution	- We distribute the tasks using a first come first serve basis. - Outcomes : Safiya is responsible for designing the ground floor, Dayang is responsible for designing General purpose lab, Farra is responsible for designing Cisco lab and lastly, Harini is responsible for designing Embedded lab.	All members
5.	Estimate the scale	- We agreed to use the 1cm = 1m scale.	All members
6.	Meeting ended	- At 10.30 p.m., Dayang ends the meeting after all discussions have been done.	Dayang